SYRIA TEL COMMUNICATION CHURN PREDICTION

BY KEVIN KISENGU

GOAL

- Project Goal: Predict customer churn to enable proactive retention strategies
- Tools: Python, Pandas, Scikit-learn, SMOTE, Matplotlib
- Target Audience: Telecom business managers

PROJECT OVERVIEW

- we want to identify “at-risk” customers before they leave since every customer who leaves costs syriatel money.
- This can be achieved by pondering the question, “Are there any predictable patterns in customer behaviour?”

DATA ANALYSIS/APPROACH

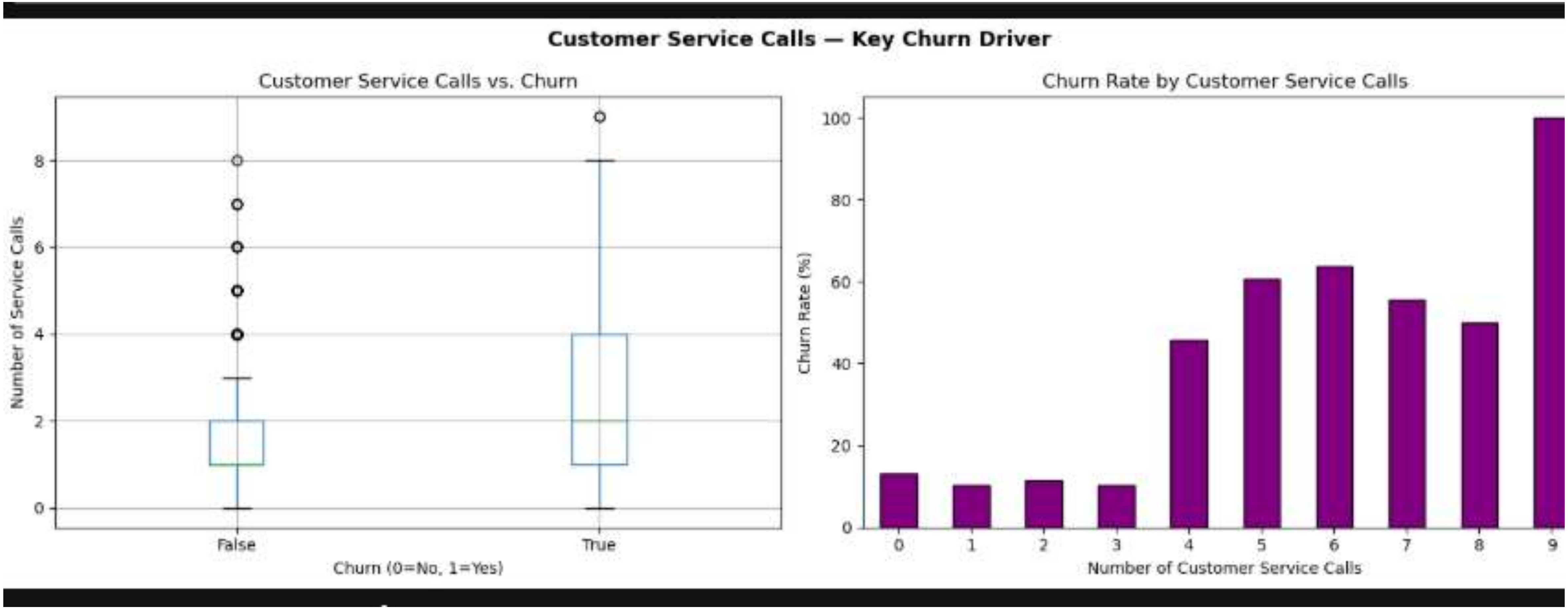
.Data cleaning and preprocessing of customer data

.Handling class imbalance using SMOTE

- We Built two different prediction models using the syriatel churn dataset
  (LOGISTIC REGRESSION and DECISION TREE)
- Trained on 80% of customers and tested on the remaining 20%.

  KEYWORDS
- roc-auc - a measure of how well the model seperates leavers from stayers.
- recall - of all customers who actually left,how many did we catch?
- precision - of customers flagged as “high risk” , how many actually left?

EDA 1 BY CUSTOMER SERVICE PLANS

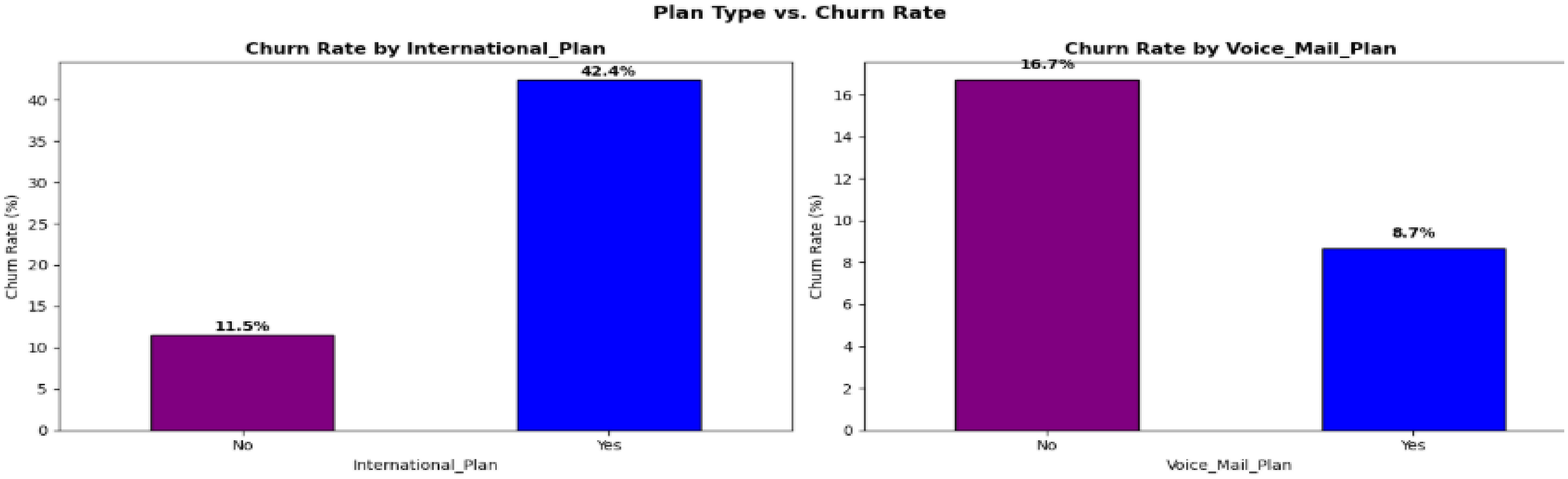


EDA 1 BY CUSTOMER SERVICE PLANS: cont

- Interpretation: Customers with an international plan churn at nearly 3x the rate of those without.

- Customers without a voicemail plan churn more than twice as often, suggesting voicemail may increase engagement and loyalty.

EDA 2. CHURN RATE BY INTERNATIONAL PLAN



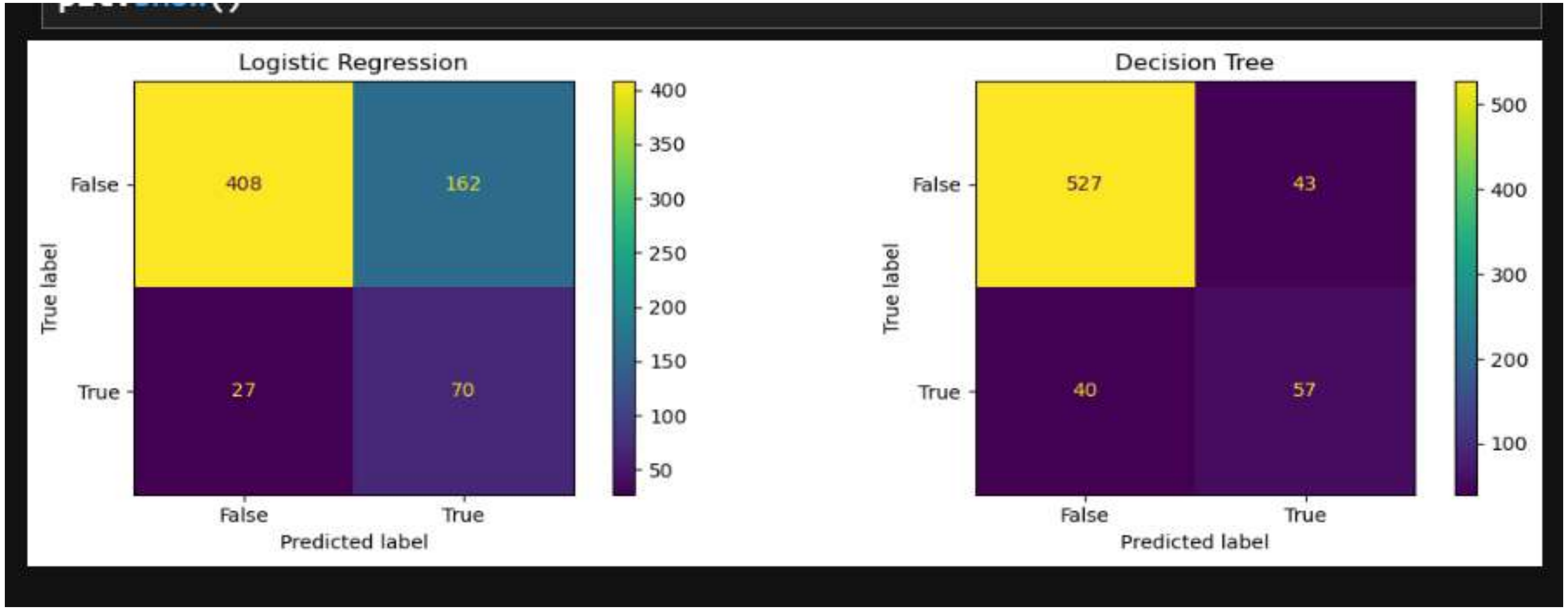
Interpretation: Churn rate spikes sharply after 4+ customer service calls (over 60% churn for 5+ calls). This is the strongest single indicator of churn and should trigger immediate retention outreach.

BUSINESS FINDINGS

- we found factors that increase churn risk:
- 1. high number of customer service calls (greater than 3) = risk
- 2. high daytime charges meant customers felt overcharged
- 3. having an international plan
- we also found factors that reduce churn risk
- 1. having a voicemail plan
- 2.higher evening usage

BUSINESS FINDINGS:cont

- Both models work well, but logistic regression is more reliable for daily predictions since it correctly identifies 8/10 customers who will churn and gives accurate risk scores making it more robust for daily use in comparison to its decision tree counter-part.



BUSINESS RECOMMENDATIONS

ACTION	TARGET
call retention team	customers with more than 3 service calls in a month
weekly risk report	all active customers scored by the logistic regression model(immediate outreach)

NEXT STEPS

- A simple dashboard should be created to update churn risk on a weekly basis(excel would suffice)